

Complex Frequency, Local Synchronization & Transient Slack Capability

PSCC 2026 Tutorial

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Low-inertia power systems

In one line

One complex quantity $\bar{\eta} = \rho + j\omega$ fuses the rate of change of a bus voltage's *magnitude* (ρ) and its *angular frequency* (ω), giving a single language for the dynamics, stability and synchronization of converter-dominated grids.

Key ideas

- **Complex frequency:** with $\bar{v} = v e^{u+j\theta}$ ($u = \ln v$),

$$\bar{\eta} = \frac{d}{dt}(u + j\theta) = \rho + j\omega, \quad \rho = \dot{v}/v, \quad \omega = \dot{\theta}.$$

- **Complex power** $\bar{s} = \bar{v} \circ \bar{i}^* = p + jq$, with rate of change (RoCoP) $\bar{s}' = \bar{\eta} \bar{s} + \bar{v} \circ \bar{i}'^*$.
- **Local synchronization:** a grid-side view of synchronism — device is synchronized when $\rho \rightarrow 0$ and $\omega \rightarrow \omega_r$ (bounded/asymptotic, BLS/ALS).
- **Transient Slack Capability (TSC):** what a device needs to sustain the grid after a large disturbance; realised by the **Dual grid-forming converter**.

Concept map

Park vectors $\rightarrow \bar{\eta}$ (complex freq.) $\rightarrow$ RoCoP $\bar{s}' \rightarrow$ device component $\bar{\chi} = \bar{\xi} - \bar{\eta} \rightarrow$ local sync (BLS/ALS) $\rightarrow$ TSC $\rightarrow$ Dual-GFM application.

Worked takeaways (verified)

- $\bar{v}' = \bar{v} \circ \bar{\eta}$; $\bar{i}' = \bar{\eta} \bar{i}$ (constant $\bar{Y}$).
- RoCoP: $\bar{s}' = \bar{\eta} \bar{s} + \bar{v} \circ \bar{i}'^*$.

- General component: $\bar{i}' = \bar{i}(\bar{\chi} + \bar{\eta}) \Rightarrow \bar{\chi} = \bar{\xi} - \bar{\eta}$.

Why ρ, ω combine: both are $[\text{time}]^{-1}$ (see `verify.md`).

Prerequisites (your profile)

Park/dq transform & Hadamard $\circ$ (`seen`); complex power (`followed`); target for $\bar{\eta}$ itself: `followed` $\rightarrow$ `intuitive`.

Where it fits

Milano's programme: *Complex Frequency* (TPWRS 2022, arXiv:2105.07769) $\rightarrow$ *Local Synchronization* (Ponce & Milano 2025) $\rightarrow$ *Dual-GFM & Transient Slack Capability* (2025).

Relevance to my work

Speaks to `effective_inertia`: $\bar{\eta} = \rho + j\omega$ and RoCoP split power-rate into voltage (ρ) and frequency (ω) parts — a language for the fast-response *offset* beyond H_{load} and for $r = H_{\text{eff}}/H_{\text{load}}$ (Milano's stored-vs-delivered gap). $\rho = \dot{v}/v$ is exactly your CMLD voltage-response / feeder question; $\bar{\chi} = \bar{\xi} - \bar{\eta} +$ local sync formalise the motor slip-torque coupling. Continuous framing — *not* `pv_trip`'s discrete 49.5/49 Hz switching.